



High Bandwidth Mode in Overwatch

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ABSTRACT

Interactive applications such as multiplayer online games require user interactions to be handled responsively, with the smallest amount of delay possible, because users must respond rapidly to the actions of other users in these types of games. A delay of even 50ms has been determined to make some multiplayer games to be frustrating and unplayable in previous investigations. However, there has not been sufficient research conducted to determine an acceptable threshold of delay for the playability of more modern games, as much of the data collected has been from multiplayer games created in the 1990s and early 2000s.

This research focused on one of the most popular modern team-based multiplayer games, known as Overwatch, which has made the claim that its relatively new high bandwidth mode will drastically increase the quality of experience for players. It also claims to group players with similar bandwidth. It is a first-person shooter, where two teams of six players compete against each other to fulfill various objectives, using player characters with different types of weapons, including hit-scan and projectile-based ones.

MATERIALS & METHODS

Released in 2016, *Overwatch* is a team-based online multiplayer first-person shooter game, developed and published by Blizzard Entertainment. They claimed their high bandwidth mode would greatly improve the quality of experience for Overwatch players.



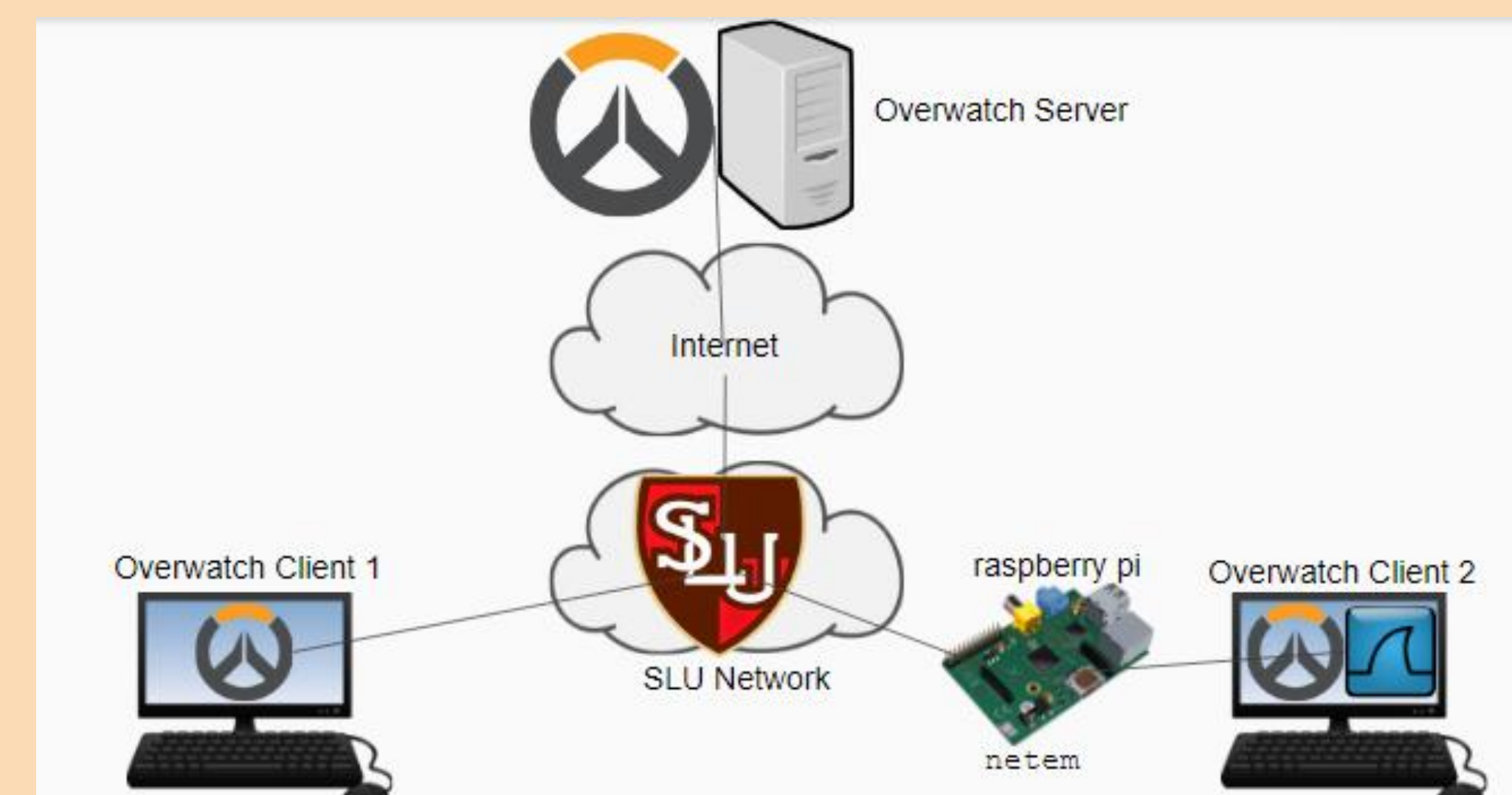
Fig. 1: the Overwatch interface

Bandwidth refers to the volume of information per unit of time that a computer is capable of sending and receiving. Higher bandwidth connections can move more data faster than lower bandwidth connections. An interactive game like Overwatch is time-sensitive, and needs info to be sent almost instantaneously.

If a packet of data takes longer to arrive than was expected, that is called **delay**. Differences in the amount of delay experienced from packet to packet is known as **jitter**. And finally, **packet loss** refers to packets failing to reach their destination after being sent.

To simulate playing Overwatch with poor network connections, this research utilized a package called **netem** to introduce delay, jitter, and loss of packets. Then, the traffic between the client and the Overwatch server was recorded with a software package known as **Wireshark**.

Fig. 2: Software setup



RESULTS

Figure 3 is a cumulative distribution function (CDF) of the bandwidth of received packets of a typical 12-minute game of Overwatch in both low and high bandwidth mode, taken every 10ms.

- **High bandwidth mode median bandwidth: 250Kbps**
- **Low bandwidth mode median bandwidth: 100Kbps**
- **2.5 times greater capacity in high bandwidth mode**

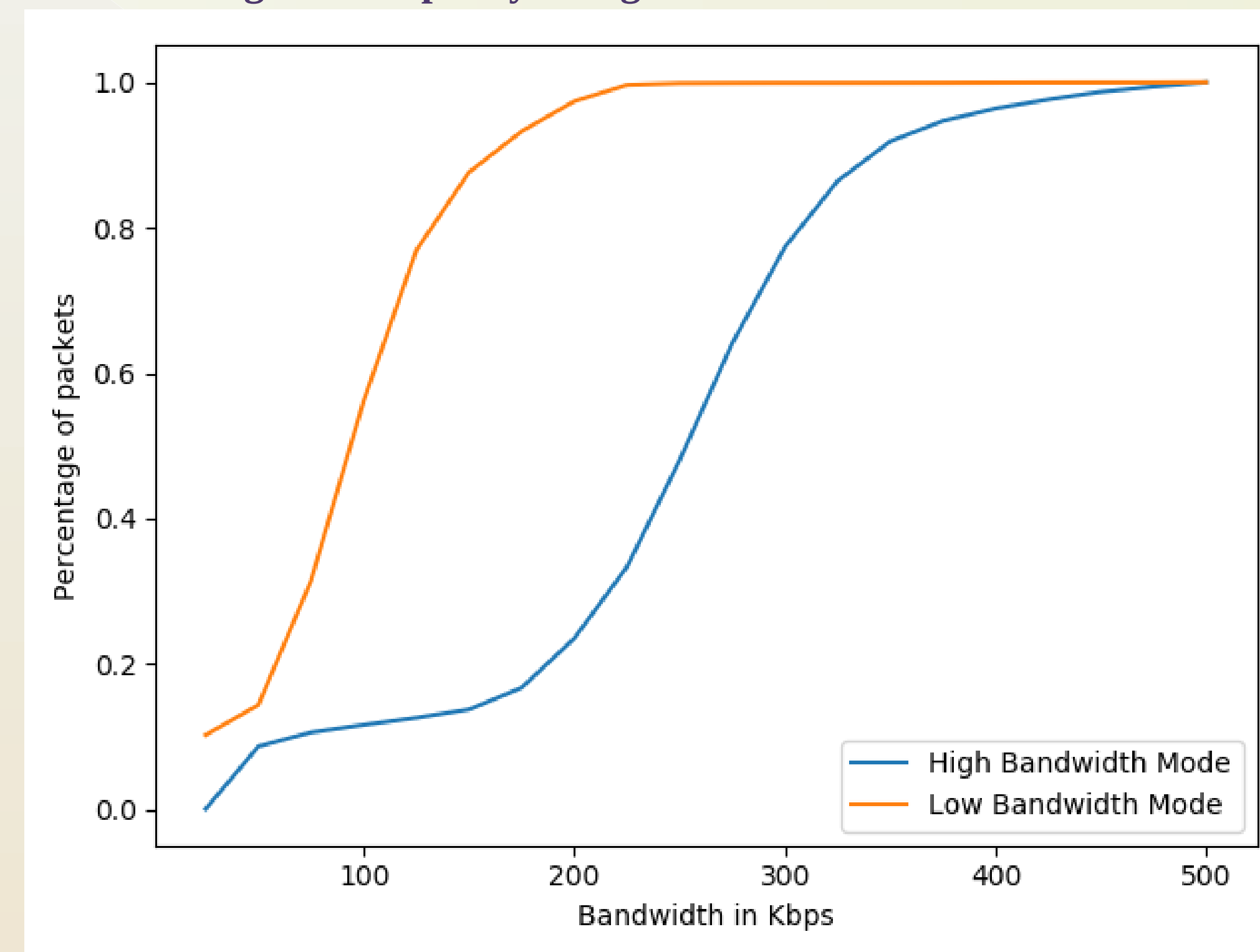


Fig. 3: CDF of bandwidth in Kbps

Figure 4 is a CDF of interpacket arrival times (how much time there is between packets arriving at their destination).

- **High bandwidth mode sends packets 2.5 times faster**
- **High bandwidth mode receives packets 4 times faster**
- **It has a higher bandwidth because it sends packets of the same size at more frequent intervals**

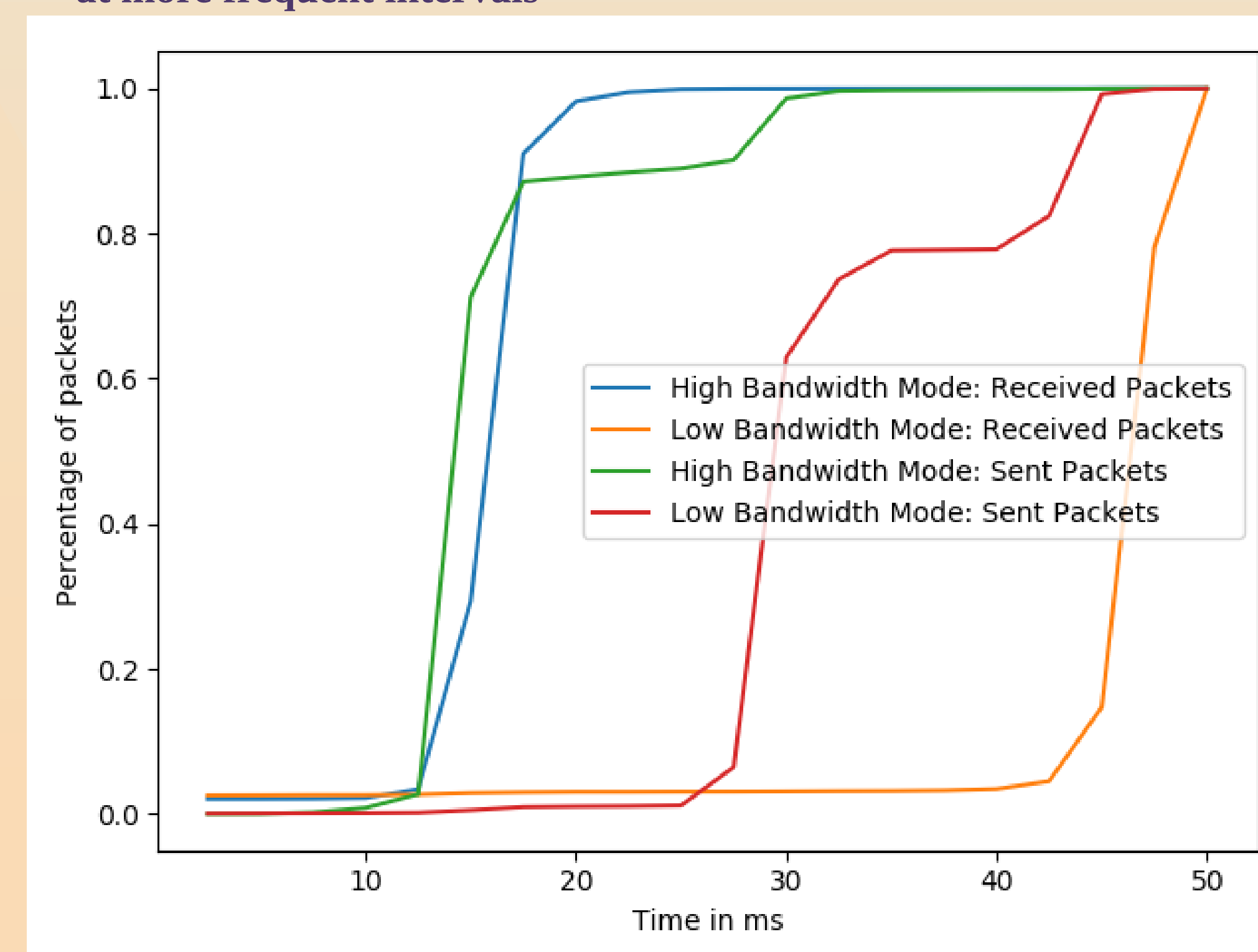


Fig. 4: CDF of interpacket arrival times

High bandwidth mode obviously gives players an edge in terms of the amount of information that they are sent and the rate at which they receive it. This allows the game to update the player's environment more frequently, and results in a smoother gameplay experience for the player.

RESULTS (cont.)

Figs. 5-8 are CDF graphs of delay, jitter, and loss conditions in high bandwidth mode which illustrate the threshold where the high bandwidth mode begins to behave like low bandwidth mode.

Fig. 5: CDF of delay

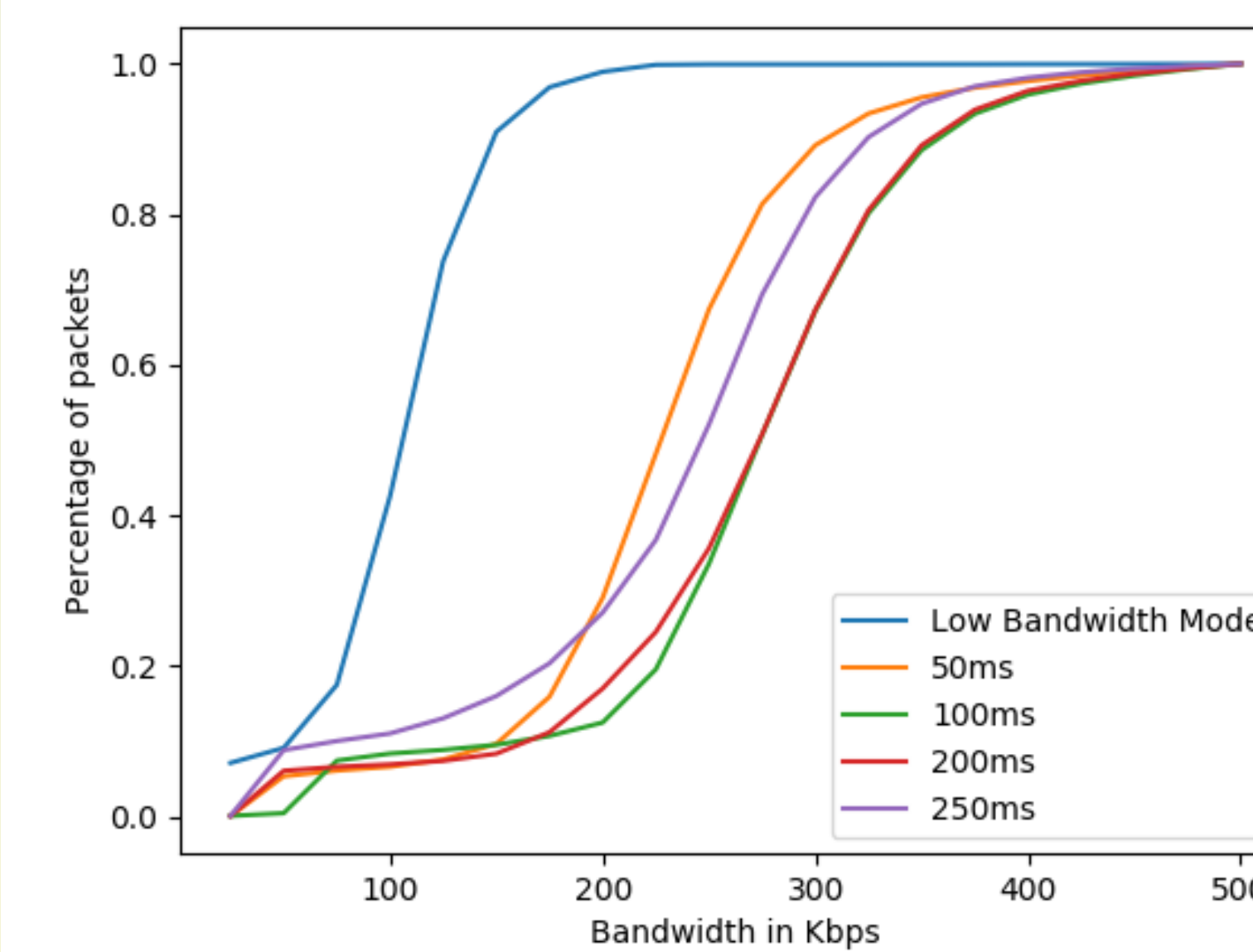


Fig. 6: CDF of uniformly distributed jitter (100ms delay)

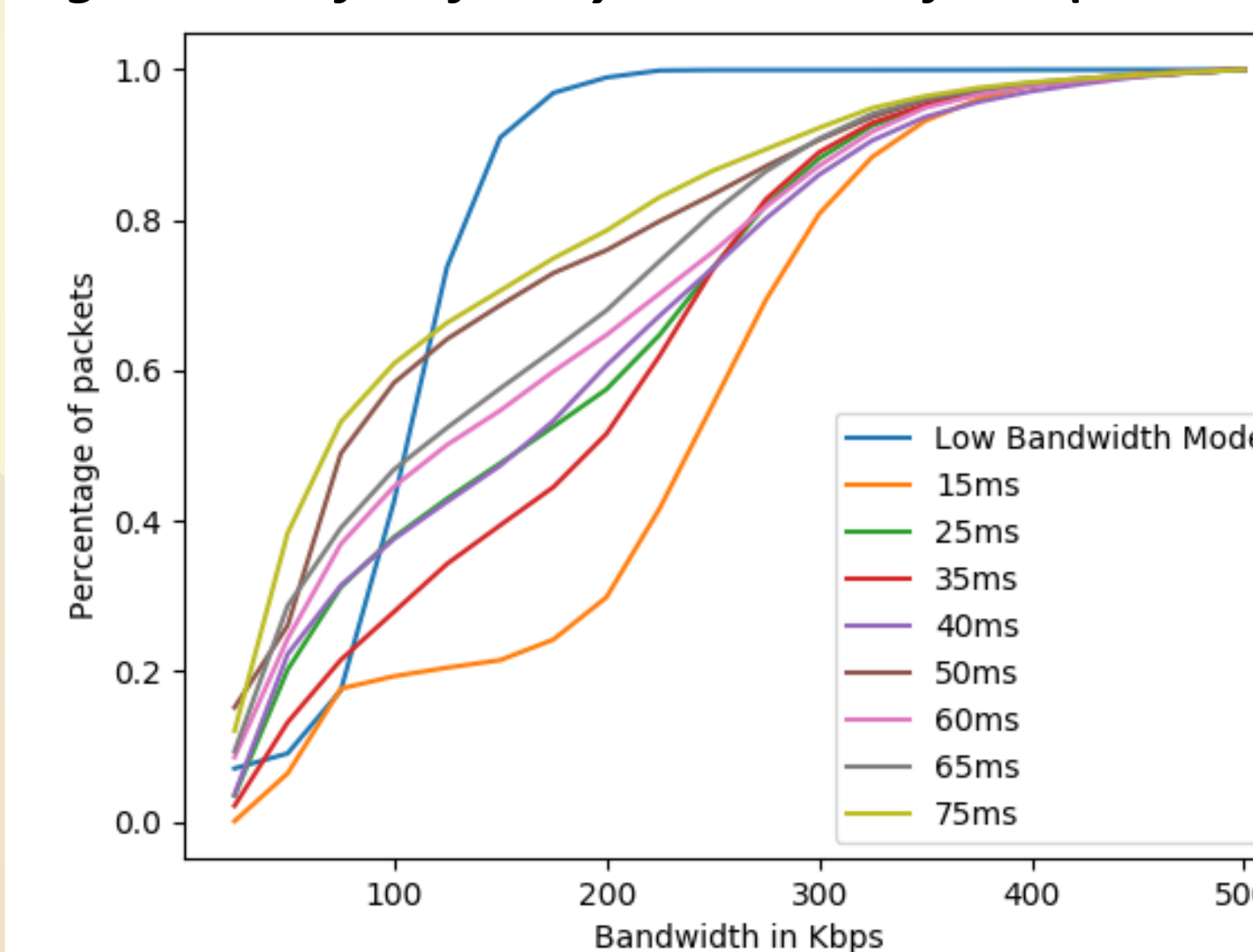


Fig. 7: CDF of normally distributed jitter (100ms delay)

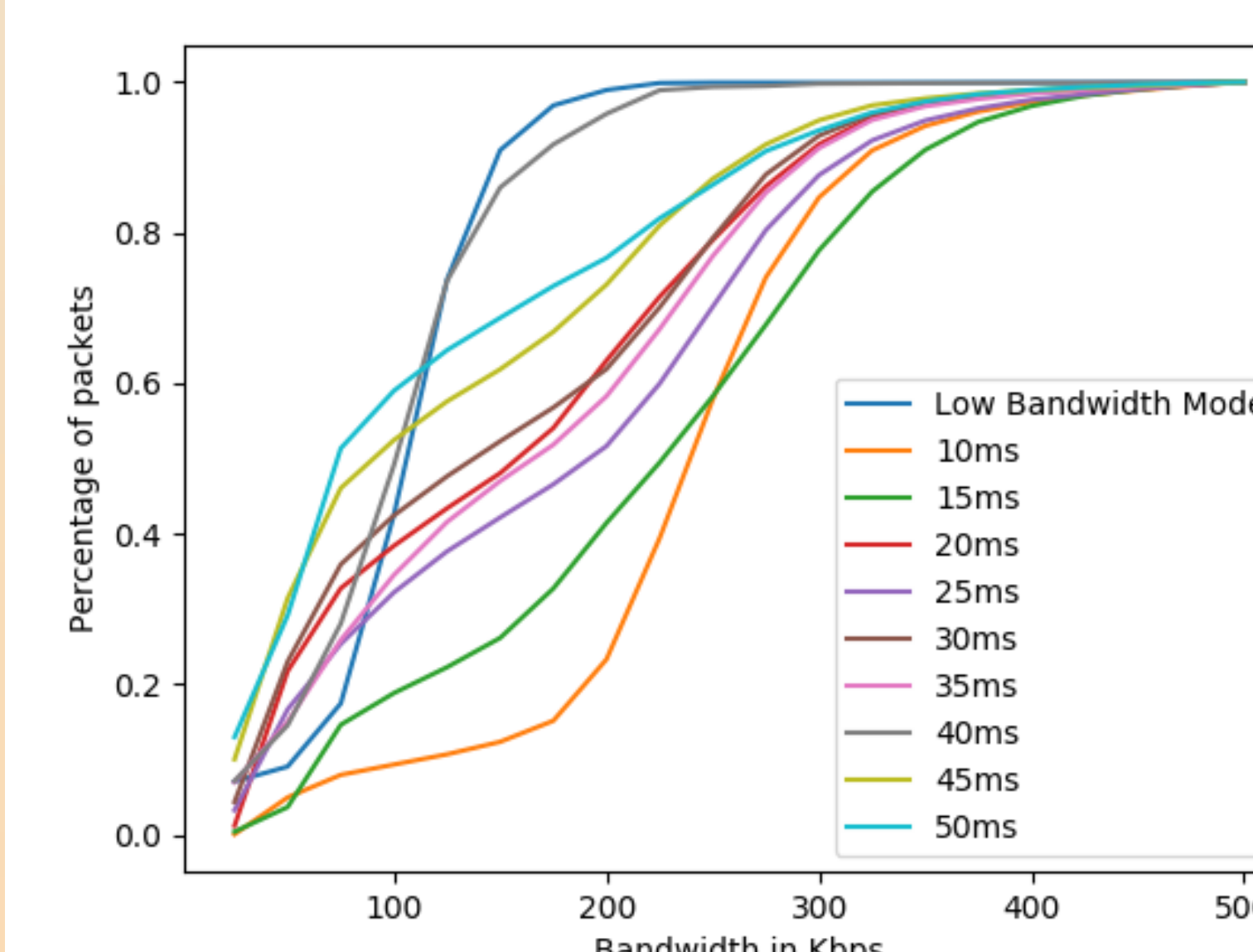
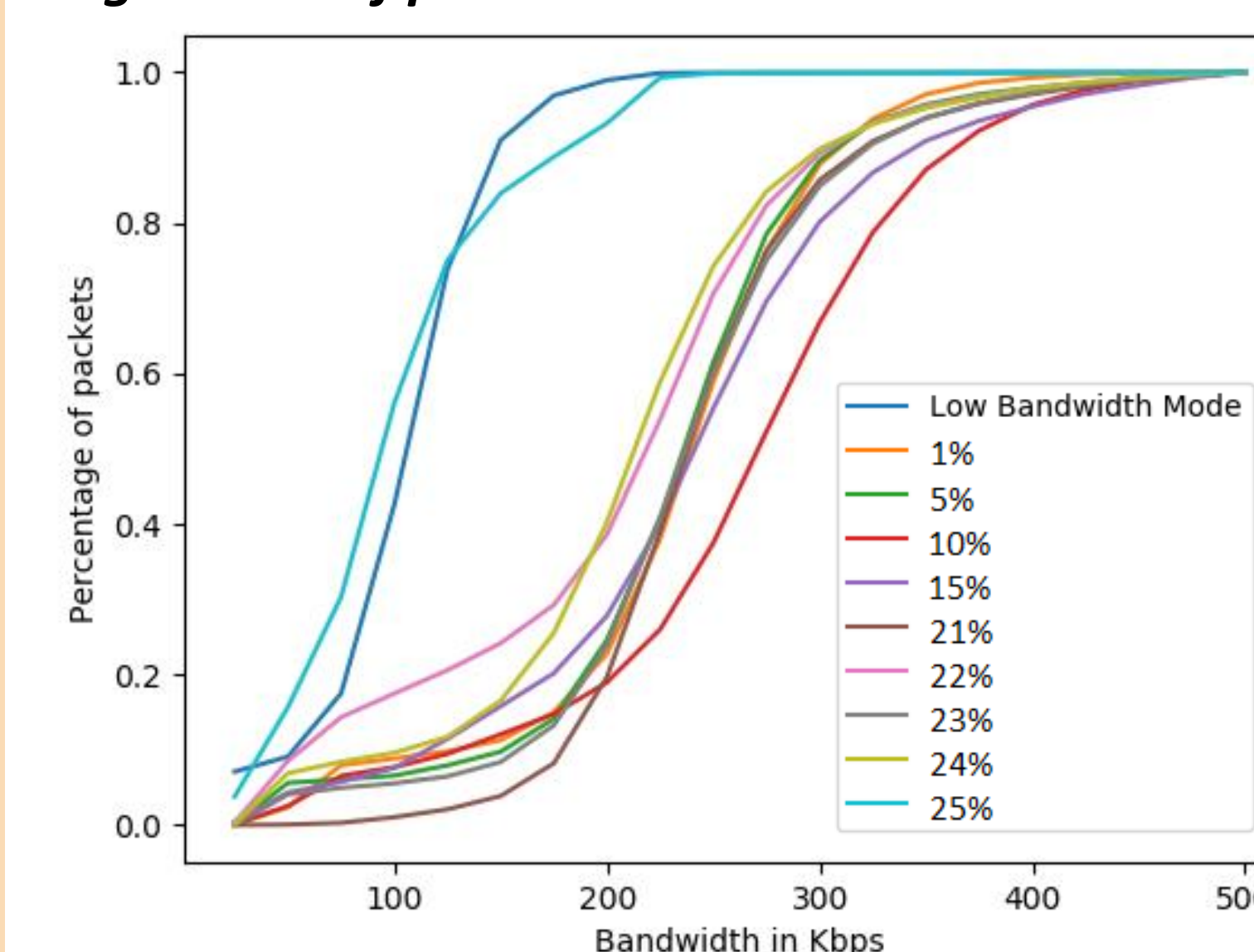


Fig. 8: CDF of packet loss



CONCLUSIONS

- **Delay with no jitter (Fig. 5):** There is clearly no effect on the median bandwidth of the received packets. In other words, **delay does not impact the performance of the high bandwidth mode**.
- **Uniformly distributed jitter (Fig. 6):** The median bandwidth begins to reach the low bandwidth mode median at about **60ms**.
- **Normally distributed jitter (Fig. 7):** Arguably more representative of how jitter would be experienced in real life (as “spikes” of delay), the threshold is closer to **45ms**.
- **Loss (Fig. 8):** At 25% packet loss, it seems to shift directly from high to low bandwidth mode, instead of the gradual shift seen in the jitter conditions. This is because until that point, the server is choosing to send more packets to make up for the fact that some are being lost in order to keep the bandwidth at the high bandwidth mode median of 250Kbps. At **25% packet loss**, it chooses to stop doing this and instead allows you to be lowered to the low bandwidth mode.

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